Explainable AI based Adversarial Examples and its Evaluation

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ABSTRACT

With the advancement of AI technology, AI can be applied to various fields. Currently, AI is used for vital decision such as medical diagnosis. Therefore, the accountability for the decision by AI has become more important. The explainable AI (XAI) is a key technology to fulfill the accountability. While AI is spreading rapidly, the vulnerability of AI system is pointed out. The adversarial examples (AE), which causes wrong decisions by AI, is one of the terrible attacks for AI. For using AI safely, AE has to be investigated thoroughly. In order to clarify mechanism and affect of AE, this paper proposes the generating method for AE which is based on XAI. Experiments prove the proposed method is superior to previous AEs.

CCS CONCEPTS

• **Machine learning**; • **Machine learning approaches**; • **Neural networks**;

KEYWORDS

security, deep neural networks, adversarial examples, explainable AI

ACM Reference Format:
Ryo Kumagai, Shu Takemoto, Yusuke Nozaki, and Masaya Yoshikawa. 2023. Explainable AI based Adversarial Examples and its Evaluation. In *2023 The 6th International Conference on Electronics, Communications and Control Engineering (ICECC 2023), March 24–26, 2023, Fukuoka, Japan.* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3592307.3592342

1 INTRODUCTION

In recent years, the deep neural networks (DNN) are used for security cameras and autonomous driving cars as AI technology, and their importance will be increased. With spreading fields of which DNN systems are applied such as real-world applications, the accountability of the AI is required. The explainable AI (XAI) [1] can fulfill accountability about its decision on the inference process of DNNs.

While AI is useful, the risk of an attack on DNN by a malicious attacker has emerged. Therefore, it is important to evaluate the threat of the attack in order to use DNN systems safely. The adversarial examples (AE) have been reported as attacks that induce malfunctions in DNNs that perform image classification [2–4]. AE causes misclassification by adding subtle perturbations to input images.

To clarify mechanism and effect of AE, this paper proposes an AE generating method. The proposed method utilizes XAI to limit the area of perturbations. The limitation achieves misclassification with less noise.

2 RELATED WORKS

This section describes Grad-CAM and FGSM which is used in the proposed method.

2.1 Grad-CAM

Class Activation Mapping (CAM) is one of the methods to explain the basis of image classification by convolutional neural network (CNN) [5]. In CAM, the feature map is multiplied by the weight of the fully connected layer, and the features of the input image are visualized by superimposing them. A feature map is the output of a convolutional layer in a CNN and contains features extracted from an image.

Since CAM is the sum of the feature maps multiplied by the weights of the fully connected layers, it has the advantage that the amount of computation is small compared to other XAI methods, such as LIME [6] and SHAP [7], which require iterative computation. However, using CAM requires that the configuration of the network architecture includes Global Average Pooling (GAP) [8]. GAP is a method to obtain the pixel average of each channel after the final layer of the convolutional layer to make the feature map one-dimensional, and it has the effect of reducing the weight parameter compared to MaxPooling.

Grad-CAM overcomes the CAM's limitations of network architecture [9]. Grad-CAM uses the gradient information of the final layer in the convolutional layer instead of the weight of the fully connected layer. Since Grad-CAM can be used for CNN without GAP, it can be applied to more various networks than CAM. Figure 1 shows an example of Grad-CAM. When the left picture of Figure 1 is input to Grad-CAM with the label "tiger_cat", the middle is output as a heat map. As can be seen from the right, the heat map

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ICECC 2023, March 24–26, 2023, Fukuoka, Japan
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ACM ISBN 979-8-4007-0000-2/23/03…$15.00
https://doi.org/10.1145/3592307.3592342

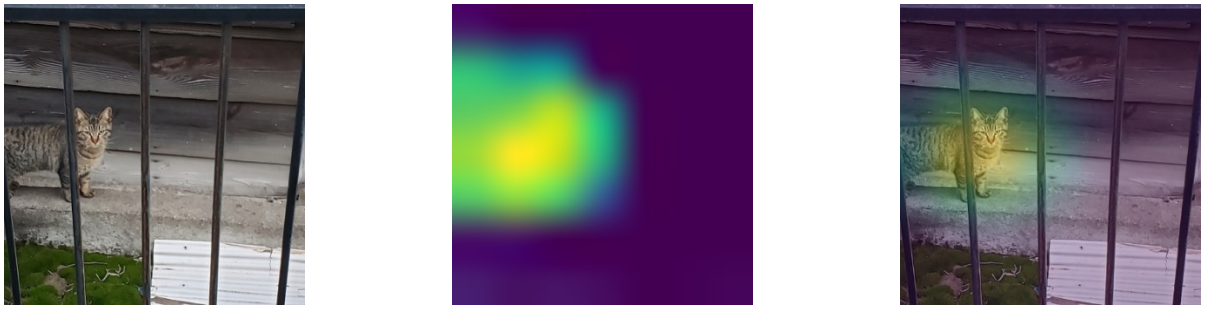

Figure 1: An example of Grad-CAM. The sample image (left), heat map by Grad-CAM (middle), and superimposed (right)

overlaid on the input image, the portion from the head to the chest of the cat has a high degree of contribution.

2.2 Fast Gradient Sign Method (FGSM)

The AE are specialised inputs generated with the purpose of causing the misclassification to the inference networks [2–4]. Normal input and AE are indistinguishable to the human eye.

One of the non-targeted AE generation methods for DNN that performs image classification is the Fast Gradient Sign Method (FGSM) [10]. FGSM uses the gradient of the input image to calculate a perturbation pattern that increases the loss of the network for each pixel of the image. An AE is generated by adding this perturbation to the input image. FGSM is expressed as

$$adv_x = x + \varepsilon * \text{sign}\left(\nabla_x J\left(\theta,\ x,\ y\right)\right)$$

where adv_x is AE, x is input, y is correct class, ε is degree of perturbation, and $\nabla_x J(\theta,\ x,\ y)$ is gradient of the error about x.

In FGSM, an image is input only once to obtain the gradient of the network, and inference is performed on it. Therefore, iterative reasoning is not required, and processing can be performed at high speed.

2.3 Jacobian Saliency Map Attack (JSMA)

Jacobian Saliency Map Attack (JSMA) is one of targeted AE generation methods [11]. In FGSM, perturbations are added to all pixels so that the error with the correct class increases, but in JSMA, a saliency map is used to assign the perturbations to pixels that have a strong influence on the target class. Add a perturbation that has a small error with the target class and a large error with the other. JSMA has a higher computational cost than FGSM because it has to calculate the gradients of all classes and iteratively adds perturbations until the score of the target class exceeds the threshold.

3 PROPOSED METHOD

FGSM can generate AEs quickly, but it only constrains the perturbation strength at a pixel and does not consider the number of changes across the image. Therefore, this study proposes two efficient AE generation methods using Grad-CAM and FGSM. By using FGSM for perturbation generation, the AE is generated with less computational effort than JSMA. The method 1 limits the number of perturbed pixels, and the method 2 limits the perturbation magnitude at each pixel.

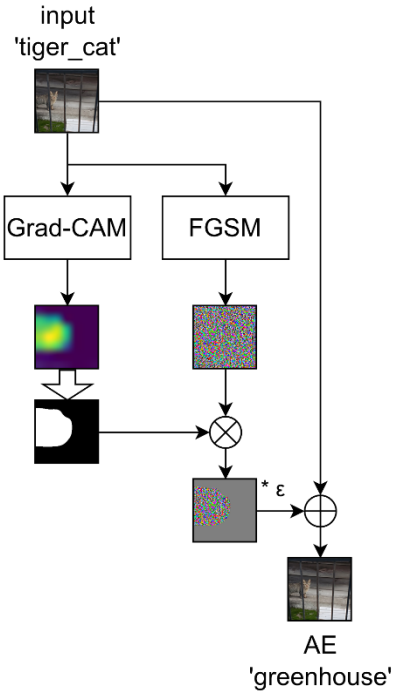

Figure 2: An outline of the proposed method 1

3.1 The Proposed Method 1

Figure 2 shows an outline of the proposed method 1. First, a perturbation pattern for the input image is generated by FGSM. Next, make the heat map using Grad-CAM. It visualizes the position where the network focused on in the input image. Then, the heat map is binarized by the threshold to prepare a mask pattern. The shape of the heat map is height x width, with each pixel ranging from [0, 255]. Therefore, a mask pattern is created by assigning 1 to pixel values above the threshold and 0 to values less than the threshold. Furthermore, by multiplying the perturbation pattern by the mask pattern, a pattern is generated in which the perturbation added to the unimportant areas in the image is 0. Finally, AE is generated by adding this pattern to the input image.

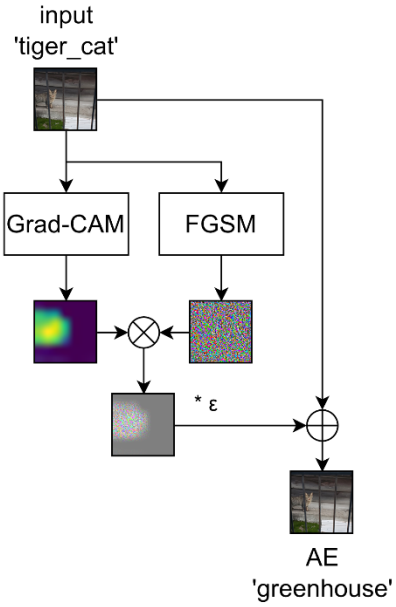

Figure 3: An outline of the proposed method 2

In this method, the perturbation is applied only to the important points in the image, so the number of pixels that change over the entire image is reduced. Therefore, the L_0 norm of noise can be reduced than FGSM.

3.2 The Proposed Method 2

An overview of the proposed method 2 is shown in Figure 3. First, a perturbation pattern and a heat map are made by the same process as the method 1. The heat map can be regarded as a weight map regarding the importance of each pixel, so that normalize the heat map from [0, 255] to [0, 1]. Next, multiply the normalized heat map with the perturbation pattern. This pattern is strong in the parts that show the features of the image well, and the parts that do not are weak. The AE is obtained by adding the perturbation pattern obtained by this procedure to the input image.

This method adds strong perturbations to the important parts of the image and weak perturbations to the less important parts. Therefore, the L_∞ norm (infinity norm) of noise at a certain pixel in the image can be reduced.

4 EVALUATION EXPERIMENTS

In the evaluation experiment, we generate AEs by the proposed methods and attack the pre-trained network. In addition, we evaluate the qualities of the original input images and AEs. In this experiment, MobileNetV2 [12] trained by ImageNet [13] is used as a pre-trained network.

4.1 AEs by FGSM

As a baseline, AEs are generated by the conventional FGSM.

The input image is Figure 4. The inference result of this image by MobileNetV2 is "tiger_cat," and the confidence is 32.49%.

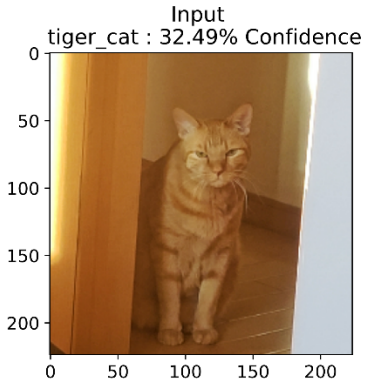

Figure 4: Original input image

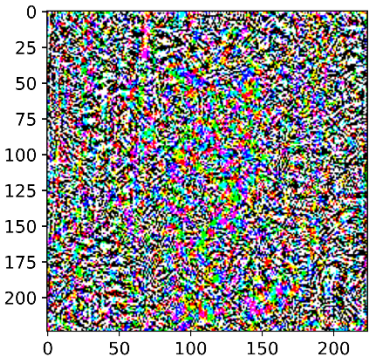

Figure 5: Perturbation pattern

Figure 5 shows the perturbation pattern obtained for the input image. Table 1 summarizes the perturbation patterns added to the input image while changing ε such as 0.01, 0.05, 0.1 and the inference results.

4.2 Generate AEs by the Proposed Method 1.

Generate AEs by the proposed method 1. The left image of Figure 6 shows the heat map obtained by applying Grad-CAM to the input image in Figure 4. The heat map is a matrix with values [0, 255]. The middle of Figure 6 shows the result of generating a mask pattern with a threshold of 96, and the right shows the result of multiplying the perturbation pattern by the mask pattern. The threshold was set so that the values of 0 and 1 in the pattern were visually halved.

Table 2 summarizes the AEs generated while changing ε such as 0.01, 0.05, 0.1 and the inference results.

4.3 Generate AEs by the Proposed Method 2.

Generate AEs by the proposed method 2. The heat map obtained by the same process as the proposed method 1 (the left image of Figure 6) is normalized to [0, 1]. Figure 7 shows the perturbation pattern in Figure 5 multiplied by this.

Table 1: AEs generated by FGSM

	0.01	0.05	0.1
AE image			
Predicted class	carton	bath_towel	bath_towel
Confidence [%]	69.08	69.45	86.31

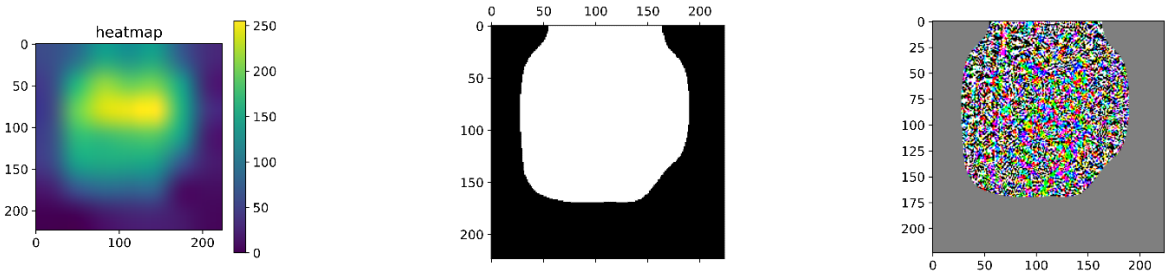

Figure 6: The Grad-CAM heatmap of the input image (left), mask pattern (middle), and the masked perturbations (right)

Table 2: AEs generated by the proposed method 1

	0.01	0.05	0.1
AE image			
Predicted class	carton	carton	shower_curtain
Confidence [%]	29.18	19.20	47.47

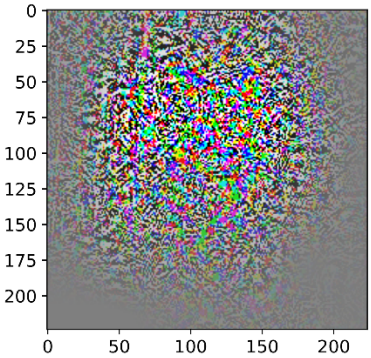

Figure 7: Product of the perturbations and the heat map

Table 3 summarizes the AEs generated while changing ε such as 0.01, 0.05, 0.1 and the inference results.

4.4 Quality Evaluation

We evaluate the quality of the AE image relative to the original input image using PSNR and SSIM [14]. PSNR is the ratio of the maximum pixel value of an image to noise, and SSIM is the index of structural similarity between two images. In both cases, the larger the number, the more similar the two samples are.

The PSNR of image x and y is defined as

$$PSNR = 10\log_{10}\frac{255^2}{MSE}$$

$$MSE = \frac{1}{3N}\sum^{3}\sum_{i=0}^{N}(x_i - y_i)^2,$$

where N is the total number of pixels in the image.

SSIM defined as

$$SSIM = \frac{(2\mu_x\mu_y + C_1)\,(2\sigma_{xy} + C_2)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)},$$

Table 3: AEs generated by the proposed method 2

	0.01	0.05	0.1
AE image			
Predicted class	carton	carton	Bath_towel
Confidence [%]	52.08	71.33	55.83

Table 4: PSNR and SSIM between the original image and AEs

method		PSNR [dB]	SSIM
FGSM	0.01	46.3155	0.9851
FGSM	0.05	33.4263	0.7865
FGSM	0.1	27.5278	0.5326
Proposed method 1	0.01	49.6249	0.9928
Proposed method 1	0.05	36.7172	0.8956
Proposed method 1	0.1	30.7931	0.7643
Proposed method 2	0.01	50.5968	0.9947
Proposed method 2	0.05	38.7274	0.9309
Proposed method 2	0.1	32.7868	0.8118

where μ is the mean and σ is the variance of each image, and C_1 and C_2 are constants.

The Table 4 summarizes the PSNR and SSIM results for the input image and the generated AEs.

In addition, generate AE for 100 images from ImageNet [13] and calculate the average of PSNR, SSIM and attack success ratio. Table 5 summarizes these.

5 DISCUSSION

Table 1, Table 2 and Table 5 show that AE attacks are successful even when the perturbation region is restricted. From this, it is considered that the proposed method 1 can reduce the L_0 norm of noise without significantly lowering the attack success rate. Also, from Table 1, Table 3 and Table 5, we can see that the attack was successful with the AE generated by limiting the strength of the perturbation. From this, it is considered that even if the L_∞ norm of the noise is reduced by the proposed method, the success rate of the attack does not decrease significantly.

In addition, Table 4 and Table 5 shows the AEs by the proposed methods can suppress image degradation more than the AEs by the conventional FGSM in all cases. Therefore, by using the proposed methods, it is possible to generate AE that is less perceptible.

6 CONCLUSION

This study proposed AE generation methods using XAI technology. The proposed method reduces computational cost for generating AE in comparison with JSMA. It also reduces adding noise by using image heat map obtained by Grad-CAM and perturbation by FGSM. Evaluation experiments evaluated the attack accuracy and the degree of image degradation for the generated AE. As a result, it was shown that the noise norm can be reduced without reducing the attack accuracy by using the proposed methods. Furthermore, the AE by the proposed methods could suppress the deterioration of the original input image more than the conventional AE by FGSM.

Future works include to modify the proposed method for applying it to the targeted attack.

ACKNOWLEDGMENTS

This work was supported by JSPS KAKENHI Grant Number JP19K11976.

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Table 5: Average of PSNR and SSIM, and attack success ratio of 100 AEs

method		PSNR [dB]	SSIM	Attack success ratio [%]
FGSM	0.01	45.7511	0.9917	89
FGSM	0.05	33.9499	0.8861	93
FGSM	0.1	28.3614	0.7354	95
Proposed method 1	0.01	47.7433	0.9951	87
Proposed method 1	0.05	36.3200	0.9347	92
Proposed method 1	0.1	30.7473	0.8453	94
Proposed method 2	0.01	49.6159	0.9976	87
Proposed method 2	0.05	39.2130	0.9661	93
Proposed method 2	0.1	33.3980	0.9029	95

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